

A. Labor Mix

The table below provides the estimated composition of the annual hours by Labor Category. The Offeror shall propose the Level of Effort provided below. Offerors' proposals may not deviate from the total number of hours included in Section B.

Labor Category	CLIN 2000 Annual Labor Hours	CLIN 2100 Annual Labor Hours	CLIN 2200 Annual Labor Hours	CLIN, 2300 Annual Labor Hours	CLIN, 2400 Annual Labor Hours	Total Hours
PROGRAM MANAGER (K)	1,920	1,920	1,920	1,920	1,920	9,600
LEAD PRINCIPAL SYST EMS ENGINEER (K)	11,520	11,520	11,520	11,520	11,520	57,600
LEAD SENIOR SYSTEMS ENGINEER (K)	19,200	19,200	19,200	19,200	19,200	96,000
PRINCIPAL SYSTEMS ENGINEER	49,920	49,920	49,920	49,920	48,000	247,680
SENIOR SYSTEMS ENGINEER	28,800	28,800	28,800	30,720	30,720	147,840
SYSTEMS ENGINEER	23,040	23,040	24,960	24,960	26,880	122,880
ENGINEER	9,600	11,520	13,440	15,360	15,360	65,280
JUNIOR ENGINEER	3,840	5,760	7,680	7,680	7,680	32,640
TOTAL	147,840	151,680	157,440	161,280	161,280	779,520
Surge (10%)	14,784	15,168	15,744	16,128	16,128	77,952
TOTAL (Including Surge)	162,624	166,848	173,184	177,408	177,408	857,472

B. Government Defined Labor Categories:

NON-KEY PERSONNEL – MINIMUM QUALIFICATIONS

In order to provide additional clarification to the Statement of Work, minimum qualifications are provided for Non-Key Personnel. The Contractor shall provide Non-Key Personnel who meet or exceed the minimum qualifications provided below. See SOW C.19, prior to charging Non-Key Personnel. During the life of the Task Order the Contractor has autonomy to manage Non-Key personnel and Non-Key categories as they deem appropriate to perform the tasking required.

1) PRINCIPAL SYSTEMS ENGINEER

Twenty (20) years of General Surface Navy Systems engineering experience with at least seven (7) years of specialized experience. Specialized experience is defined as significant systems engineering responsibilities in one or more of the following areas:

- (a) Combat Systems Cyber Security
- (b) Ships Self Defense System (SSDS)
- (c) Combined Integrated Air and Missile Defense Antisubmarine Warfare Trainer (CIAT)
- (d) Littoral Combat Ship (LCS) class frigate
- (e) Aegis Integrated Combat Systems
- (f) DDG-1000 Zumwalt class destroyer
- (g) Track Management
- (h) Aegis Ballistic Missile Defense (ABMD)
- (i) Integrated Combat System (ICS)
- (j) Weapon/Combat Systems Experience
- (k) Naval or Joint Interoperability
- (l) Common Aviation Command and Control System (CAC2S)

2) SENIOR SYSTEMS ENGINEER

Fifteen (15) years of General Surface Navy Systems engineering experience with at least seven (7) years of specialized experience. Specialized experience is defined as significant systems engineering responsibilities in one or more of the following areas:

- (a) Combat Systems Cyber Security
- (b) SSDS
- (c) CIAT
- (d) LCS class frigate
- (e) Aegis Integrated Combat Systems
- (f) DDG-1000 Zumwalt class destroyer
- (g) Track Management
- (h) ABMD
- (i) ICS
- (j) Weapon/Combat Systems Experience
- (k) Naval or Joint Interoperability
- (l) CAC2S

3) SYSTEMS ENGINEER

Ten (10) years of General Surface Navy Systems engineering experience with at least five (5) years of specialized experience. Specialized experience is defined as significant systems engineering responsibilities in one or more of the following areas:

- a) Engineering investigation and resolution of combat system operational issues resulting in system performance improvements
- (b) The development of methodologies, models, or simulations for conducting engineering assessment to Weapon System or Combat System performance; combat system performance data analysis, simulation/development and maintenance
- (c) Development/maintenance of combat system element-level quality assurance and/or configuration management procedures
- (d) Computer program definition, design, development, test and multi-element integration
- (e) Combat system element-level and associated computer program documentation development/maintenance
- (f) Multi-element test plans/procedures development/maintenance
- (g) Exploration of new engineering war fighting capabilities for Carriers, Amphibious and/or Aegis ships
- (h) The design engineering associated with ICS, SSDS, CAC2S, ABMD, AWS and/or ACS enhancements
- (i) Human systems integration engineering
- (j) System Safety
- (k) Naval or Joint Interoperability

4) ENGINEER

Five (5) years of General Surface Navy Systems engineering or operational experience.

5) JUNIOR ENGINEER

One (1) year of General Surface Navy Systems engineering or equivalent educational experience.

C. Geographic Distribution

For proposal purposes, the Contractor shall assume personnel will be located in the Contractor's facility except as noted in SOW C.7.

D. Key Personnel Qualifications

To perform the requirements of the Statement of Work (SOW), the Government desires Key Personnel with the appropriate experience and professional qualifications. Key Personnel qualification levels are considered to be desired for those individuals whose resumes are submitted for evaluation with the proposal. All individuals performing under the key personnel labor categories are considered key. Resumes for any replacement of key personnel that are submitted following award shall have qualifications equal to or higher than the qualifications of the person to be replaced, as required by the clause entitled C-237-H002 - Substitution of Key Personnel. Following award, the qualification levels are considered to be minimums for any growth beyond those individuals initially proposed.

All experience may be obtained concurrently for each labor category. The experience listed for each Key Labor Category should be directly related to the tasks and programs listed in the SOW.

All Key resumes must be reviewed by the Contract Specialist and Contracting Officer's Representative (COR) before the Contracting Officer approves direct charges to the contract.

The solicitation requires seventeen (17) Key Personnel positions, each of which will be filled by a single FTE. All Key Personnel positions shall be considered senior. The following table lists these Key Personnel positions, as well as the Government's desired attributes for each:

Key Personnel Position	Desired Attributes
PROGRAM MANAGER 1 FTE	Twenty-five (25) years of General Surface Navy Combat Systems engineering experience with at least seven (7) years specialized experience, with assignments in engineering development, test and evaluation, and lifetime support engineering phases, and at least five (5) years project management experience for a major tactical system employing embedded computer technology.
LEAD PRINCIPAL SYSTEMS ENGINEER 6 FTE	Twenty-five (25) years of General Surface Navy Combat Systems engineering experience with at least Ten (10) years of specialized experience. Specialized experience is defined as significant systems engineering responsibilities in one or more of the following areas: (a) Combat Systems Cyber Security (b) SSDS (c) CIAT (d) LCS class frigate (e) Aegis Integrated Combat Systems (f) DDG-1000 Zumwalt class destroyer (g) Track Management (h) ABMD (i) ICS (j) Weapon/Combat Systems Experience (k) Naval or Joint Interoperability (l) CAC2S

**LEAD SENIOR
SYSTEMS
ENGINEER**

10 FTE

Twenty (20) years of General Surface Navy Systems engineering experience with at least ten (10) years of specialized experience. Specialized experience is defined as significant systems engineering responsibilities in one or more of the following areas:

- (a) Combat Systems Cyber Security
- (b) SSDS
- (c) CIAT
- (d) LCS class frigate
- (e) Aegis Integrated Combat Systems
- (f) DDG-1000 Zumwalt class destroyer
- (g) Track Management
- (h) ABMD
- (i) ICS
- (j) Weapon/Combat Systems Experience
- (k) Naval or Joint Interoperability
- (l) CAC2S